SURVEY REPORT ON BLOCKCHAIN TYPES AND USE CASES

Blockchain Technology: Types, Applications, Benefits and Challenges

1. Abstract

Blockchain is a distributed digital ledger that records transactions across a network of computers. Records are grouped into blocks and linked using cryptographic techniques, making unauthorized alteration difficult without detection. This report surveys the main types of blockchain networks and practical use cases in finance, supply chains, healthcare, digital identity, government, and other sectors. It also summarizes benefits, limitations, and factors to consider when selecting a blockchain platform.

2. Objectives

- To understand the basic concept and working principles of blockchain technology.
- To identify and compare the major types of blockchain networks.
- To study real-world and potential applications across different industries.
- To examine the advantages, challenges, and suitability of blockchain for different problems.

3. Introduction

A blockchain is a shared ledger maintained by multiple participating computers, often called nodes. Transactions are validated according to agreed rules and recorded in a sequence of blocks. Cryptographic hashes connect the blocks, while a consensus mechanism helps participants agree on the accepted state of the ledger. Depending on the network, participants may be able to read records, submit transactions, validate blocks, or administer access permissions.

Blockchain is not automatically the best solution for every database problem. It is most useful when several parties need to share records and want a tamper-evident history without relying entirely on one organization to maintain the data. The choice of network depends on governance, privacy, performance, cost, and the level of decentralization required.

4. Types of Blockchain

Blockchain networks are commonly grouped into four broad categories. In practice, some systems combine features from more than one category.

Type	Access and Control	Typical Uses	Key Consideration
Public	Generally open to anyone to read and, depending on the network, submit transactions or participate in validation.	Cryptocurrencies, open decentralized applications, tokenized assets.	Open verification; fees, speed, and privacy vary.
Private	Controlled by one organization; access and permissions are restricted.	Internal audit trails, enterprise records, permissioned workflows.	More control and privacy, less decentralization.
Consortium / Federated	Governed by a group of selected organizations sharing validation or administration.	Interbank processes, multi-company supply chains, industry networks.	Needs shared governance and member agreement.
Hybrid	Combines restricted/private components with selected public verification or visibility.	Confidential business records with public proofs or selective disclosure.	Access design can be more complex.

4.1 Public Blockchain

A public blockchain is open and usually allows broad participation. Transactions can often be viewed by anyone, and participants follow a shared consensus protocol. Public networks are useful when openness and independent verification matter. Transaction fees, confirmation time, and public visibility of transaction metadata can be concerns.

4.2 Private Blockchain

A private blockchain restricts participation to authorized users and is typically managed by one organization. It can support controlled access, faster processing, and confidentiality for internal workflows. Its trust model depends more heavily on the organization operating it, so it may provide fewer decentralization benefits than a public network.

4.3 Consortium or Federated Blockchain

A consortium blockchain is governed by several known organizations rather than one owner or the general public. It can be suitable when organizations need coordinated records but do not want one participant to control the entire system. Rules are needed for membership, upgrades, dispute handling, and responsibility for errors.

4.4 Hybrid Blockchain

A hybrid blockchain combines private data or permissioned processes with selected public components. Sensitive business records may remain restricted while a cryptographic proof or summary is published to a public chain for independent verification. Careful design is needed to protect confidential information.

5. Real-World Use Cases

5.1 Banking and Financial Services

Blockchain can support digital asset transfers, settlement, shared transaction records, and automated workflows through smart contracts. It may reduce reconciliation between organizations and improve traceability. Adoption depends on regulation, privacy, interoperability, and procedures for correcting mistaken transactions.

5.2 Supply Chain and Logistics

Manufacturers, distributors, carriers, and retailers can record production, shipment, and delivery events on a shared ledger. This can improve traceability and help investigate delays or suspected counterfeit goods. Blockchain cannot guarantee that information entered at the source is true; audits and reliable data collection remain essential.

5.3 Healthcare

Blockchain may help coordinate access logs, consent records, clinical-trial records, and data exchange among healthcare organizations. Sensitive patient information should generally be stored off-chain or protected by strict access controls, while the blockchain stores references or audit evidence. Patient privacy and compliance are central.

5.4 Digital Identity and Certificates

Blockchain-based systems can help verify educational certificates, professional licences, and selected identity claims across organizations. Personal data should be minimized on-chain, and systems need processes for revocation, correction, and recovery if credentials or keys are lost.

5.5 Government and Public Records

Potential uses include tracking public procurement, maintaining tamper-evident audit trails, and verifying documents or licences. Shared records may improve accountability, but blockchain does not replace sound governance, lawful recordkeeping, or independent oversight.

5.6 Insurance

Smart contracts can automate parts of claims processing when predefined conditions are met, such as a verified travel delay or shipment event. Reliable external data sources, often called oracles, are needed to supply real-world facts. Incorrect data or contract logic can cause inappropriate outcomes.

5.7 Internet of Things (IoT) and Energy

Blockchain can provide shared records for device events, machine-to-machine transactions, or peer-to-peer energy trading. Device limitations, transaction speed, energy use of some consensus mechanisms, and integration with existing infrastructure must be considered.

5.8 Digital Assets, NFTs, and Decentralized Applications

Public blockchains can track ownership and transfers of tokens and support decentralized applications. A token does not automatically confer legal ownership of a physical item or copyright. Users also face risks from scams, software vulnerabilities, market volatility, and lost private keys.

6. Benefits of Blockchain

- Tamper-evident records: unauthorized changes are easier to detect.
- Shared source of truth: participants can consult a common transaction history.
- Traceability and auditability: authorized participants can review recorded events.
- Automation: smart contracts can execute defined actions when conditions are met.
- Reduced reconciliation: shared records may reduce repeated checking between organizations.

7. Challenges and Limitations

- Scalability and performance: some networks process fewer transactions per second than conventional databases.
- Privacy: transparent ledgers may reveal transaction patterns or sensitive metadata.
- Governance: participants need rules for upgrades, disputes, access, and responsibility.
- Costs: transaction fees, infrastructure, integration, and maintenance can be significant.
- Data quality: blockchain cannot guarantee that submitted information was accurate.
- Security and key management: stolen keys, unsafe smart contracts, and user mistakes can cause losses.
- Legal and regulatory requirements differ by jurisdiction and use case.

8. Survey Findings and Comparison

The survey indicates that blockchain type should be selected according to the problem rather than treated as a one-size-fits-all technology. Public networks suit applications that benefit from open participation and independent verification. Private networks suit controlled organizational workflows. Consortium networks are useful where multiple known parties need shared governance. Hybrid designs can balance confidentiality with selective public verification, although they may be more complex to build and operate.

Blockchain is most compelling when several parties need a consistent, auditable record and no single party should have unilateral control over the shared history. If one trusted organization already controls the process, a conventional database may be cheaper, faster, and easier to manage.

9. Conclusion

Blockchain is a distributed-ledger technology with public, private, consortium, and hybrid forms. Its potential applications span finance, supply chains, healthcare, identity, government, insurance, IoT, and digital assets. It can improve shared recordkeeping, traceability, and automation, but also introduces technical, operational, privacy, governance, and regulatory challenges. A successful implementation requires a clear business need, careful network selection, secure design, trustworthy input data, and ongoing management.

10. References

- Nakamoto, S. (2008). *Bitcoin: A Peer-to-Peer Electronic Cash System.*
- Ethereum.org. *Introduction to Ethereum* and documentation on smart contracts and decentralized applications.
- National Institute of Standards and Technology (NIST). *Blockchain Technology Overview*, NISTIR 8202 (2018).
- World Economic Forum. Publications and reports on blockchain applications in supply chains and digital identity.

Note: This is a general educational survey report. Specific deployments and legal requirements should be checked against current authoritative sources.